

AI Synthesis Documentation

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Python Final Project

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USAGE 1: Choosing phonemic transcription over orthographic regex

BEFORE:

My initial plan was to count sibilants by searching for letter patterns ("s", "sh", "ch", "z") in the spelling of each word using regular expressions. This seemed straightforward since the sibilant sounds appeared to correspond to obvious letter combinations. I had not yet considered the gap between English spelling and pronunciation.

THE PROMPT:

I asked Claude whether regex matching on letter spellings would work reliably for counting sibilants in English words, since my study depended on identifying sibilant phonemes accurately.

AFTER:

Claude pointed out that English orthography is unreliable: *sure* starts with /ʃ/ despite the "s," *rose* ends with /z/, *measure* contains /ʒ/ with no obvious orthographic cue, and *machine* has /ʃ/ from "ch" while *chemistry* has /k/. This convinced me that spelling-based counting would produce systematic errors. I switched my methodology entirely to phonemic transcription using the CMU Pronouncing Dictionary, accessed through Python's nltk library. This was a critical methodological correction — without it, my study would have produced unreliable results. I made this change before writing any analysis code.

USAGE 2: Updating the literature review with a missing bridge citation

BEFORE:

My presentation's literature review jumped from Svantesson's general sound symbolism work to Reilly's taboo language work with no explicit connecting reference. The logical bridge

between "sounds can carry meaning" and "taboo words might be sibilant-rich" was implicit rather than supported by a specific citation.

THE PROMPT:

I asked Claude whether there was a specific paper that empirically linked sibilants to negative or arousing affect, so I could anchor my hypothesis in prior empirical work rather than making the connection myself.

AFTER:

Claude searched the literature and identified Aryani, Conrad, Schmidtke, and Jacobs (2018), "Why 'piss' is ruder than 'pee'? The role of sound in affective meaning making," published in *PLOS ONE*. I read the paper and confirmed it was exactly what I needed: it demonstrates empirically that hissing sibilants raise a word's spectral center of gravity and are rated as more arousing and more negatively valenced, and it notes that George Carlin's seven taboo words all contain voiceless stops or sibilants. I added this as the central motivating citation for my hypothesis. This transformed my literature review from a loose sequence of citations into a logically structured argument.

USAGE 3: Selecting a modern source for taboo word identification

BEFORE:

I had initially planned to base my taboo word list on Janschewitz (2008), which is the classic norming study in the field. I was uncertain whether there was a more recent, methodologically updated alternative.

THE PROMPT:

I asked Claude for a more recent source of normed taboo words, ideally a peer-reviewed database that improves on Janschewitz's expert-curated approach.

AFTER:

Claude searched and identified Sulpizio et al. (2024), "Taboo language across the globe: A multi-lab study," published in *Behavior Research Methods*. The advantage over Janschewitz is methodological: Sulpizio et al. use a community-based (descriptive) nomination procedure rather than an expert-based (normative) selection, which avoids researcher bias and reflects how speakers actually treat words as taboo. I cited both Janschewitz and Sulpizio in my final

paper, noting that my final taboo word list was informed by both. This decision strengthened the methodological defensibility of my stimulus set.

USAGE 4: Operationally defining "neutral" with affective norms

BEFORE:

I knew I needed a baseline group of "neutral" English words for comparison, but I did not have a defensible operational criterion. Choosing words that simply *felt* neutral would have been arbitrary and would have weakened the study.

THE PROMPT:

I asked Claude how to identify neutral comparison words from a corpus in a way that would be methodologically rigorous and grounded in published norms.

AFTER:

Claude suggested using Warriner et al. (2013), a peer-reviewed database of valence and arousal ratings for 13,915 English lemmas, and proposed filtering for valence between 4.5 and 5.5 (true middle of the 1–9 scale) and arousal below 4.5 (low emotional activation). This excludes both positively and negatively charged words as well as emotionally arousing items, leaving only affectively flat vocabulary. I evaluated and accepted this operationalization because it relies on independent peer-reviewed ratings rather than on my own intuitions about which words "feel" neutral. I implemented these specific cutoffs in my filtering code.

USAGE 5: Designing the frequency-matching algorithm

BEFORE:

I understood conceptually that my neutral group needed to be matched to the taboo group on word length and frequency to avoid confounds, but I did not know how to implement matching in Python or how to define "closeness" mathematically between two words.

THE PROMPT:

I asked Claude how to match each taboo word to a single neutral word from the candidate pool, controlling for both phoneme length and log word-frequency simultaneously.

AFTER:

Claude proposed a greedy nearest-neighbor matching algorithm: for each taboo word,

compute a standardized Euclidean distance to every available neutral candidate over the two dimensions (length, log-frequency), select the closest one, and remove it from the pool so it cannot serve as a match for another taboo word. The distance formula weights a 2-phoneme length difference as roughly equivalent to a 1.0 log-frequency difference. I implemented this in my script and post hoc verified the matching quality: mean phoneme length was 3.86 in both groups (identical), and mean log-frequency was 2.83 vs 2.82. This level of precision demonstrates that the algorithm worked, and any observed sibilant difference cannot be attributed to length or frequency confounds.

USAGE 6: Selecting the appropriate statistical tests

BEFORE:

I knew I had two groups to compare on sibilant density, but I was uncertain which statistical test was appropriate. The options I considered included independent-samples t-test, Mann-Whitney U, paired t-test, and Wilcoxon signed-rank. I also was unsure whether to use a one-tailed or two-tailed test.

THE PROMPT:

I asked Claude which statistical test would be most appropriate given that (a) each taboo word was paired with a specific matched neutral word, (b) my sample was small ($n = 29$), (c) the measure was bounded between 0 and 1 with many zero values, and (d) my hypothesis was directional.

AFTER:

Claude recommended a paired Wilcoxon signed-rank test as the primary analysis (non-parametric, appropriate for matched pairs and non-normal data), with a paired t-test for parametric comparison, both one-tailed (since my hypothesis predicted higher sibilant density specifically in taboo words). It also recommended computing Cohen's d for paired samples to report effect size rather than relying on p-values alone, which is particularly important in small samples. I implemented Shapiro-Wilk normality tests first to justify the non-parametric choice (both groups deviated strongly from normal, $p < 0.0001$), then ran both the Wilcoxon and paired t-test as planned. The final result: Wilcoxon $W = 51.5$, one-tailed $p = 0.16$; Cohen's $d = 0.22$.

USAGE 7: Honestly interpreting a non-significant result

BEFORE:

My statistical test produced a p-value of 0.16 — not significant at the conventional $\alpha = 0.05$ level. I was uncertain how to frame this in my paper without either overclaiming a "real" effect or dismissing the study as a failure.

THE PROMPT:

I asked Claude how to honestly interpret and present a non-significant result that nonetheless showed a trend in the predicted direction with a small effect size.

AFTER:

Claude framed the result as scientifically meaningful in three ways: (1) the direction matched the prediction (taboo words were 52% denser in sibilants); (2) the effect size was small but non-zero ($d = 0.22$), which is itself an informative quantification; (3) a sample of $n = 29$ pairs is underpowered for a small effect — power analysis suggests roughly 130 pairs would be needed for 80% power. It also suggested two substantive interpretations: that sibilants may exert only a modest lexical-statistical effect, or that the true phonological signature of taboo language is broader (including voiceless stops). I adopted this framing in my Discussion section. This is a more mature scientific framing than either overclaiming or dismissing the result, and it positions my study as a useful contribution to the literature even though it did not confirm the directional hypothesis statistically.